Session 3: Git and LaTeX in VS Code

Version Control and the Integrated Research Environment

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Where We Are

Session 1: What Claude Code can do. How it works. CLAUDE.md and plan mode.

Session 2: The core workflow. Data cleaning, estimation, robustness, LaTeX tables.

Session 3: Building infrastructure around the workflow.

Today you will:

1. Learn the minimum viable git workflow for agentic coding
2. Bring your paper into the same project directory as your code
3. Use Claude Code to write, compile, and version-control a LaTeX document

Session 2 taught you to verify Claude Code's work. Today you get the tools to manage it.

Git is research insurance,
not software engineering ceremony.

You need exactly four commands.

What Are Git and GitHub?

Git is a tool that tracks every change to every file in a project.
It runs on your computer. It is free and open source.

Think of it as "track changes" for your entire project directory — not just one document, but every file: code, data descriptions, LaTeX files, everything.

GitHub is a website that hosts git projects online.

- It is where open-source code, replication packages, and increasingly papers live
- Git works entirely on your machine. GitHub is optional — it adds backup, sharing, and collaboration
- Today we only use git locally. GitHub comes in Session 4.

The Key Idea

Git lets you save snapshots of your project at any point. You can always see what changed between snapshots, and go back to any previous one.

Why Researchers Should Care About Git

Without git, you have probably done this:

- `analysis_v2.py`, `analysis_v2_final.py`, `analysis_v2_final_REAL.py`
- "Which version of the cleaning script produced these results?"
- "I changed something last week and now the code is broken"

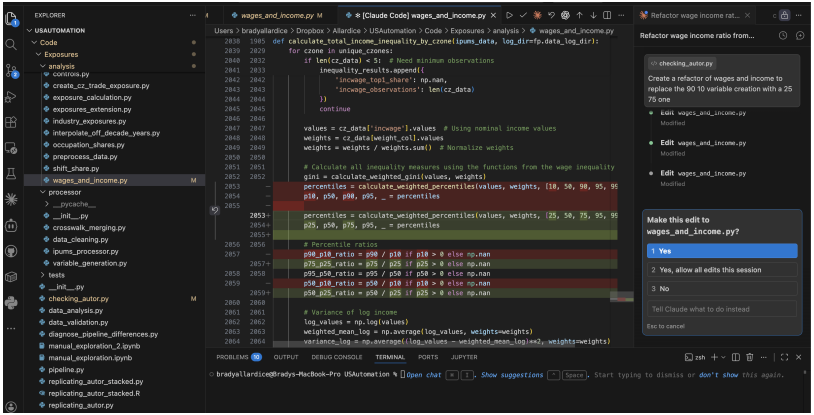
With git:

- One file, full history. Every version is recoverable.
- You can see exactly what changed, when, and why.
- You can undo any change — even from weeks ago.

Git is useful for any research project. But when you are working with an agent that modifies your files directly, it goes from useful to essential.

You Have Already Seen This

In VS Code, when Claude Code proposes a change, you see a diff view:



Red = removed, green = added. You have been reading diffs since Session 1.

Git does the same thing — but for your entire project, across *every* change, and you can undo any of them after the fact.

Why Git Matters for Agentic Coding

The problem: Claude Code modifies your files directly.

- It might change a file you did not ask it to change
- It might overwrite working code with something broken
- It might make 10 changes when you only wanted 3

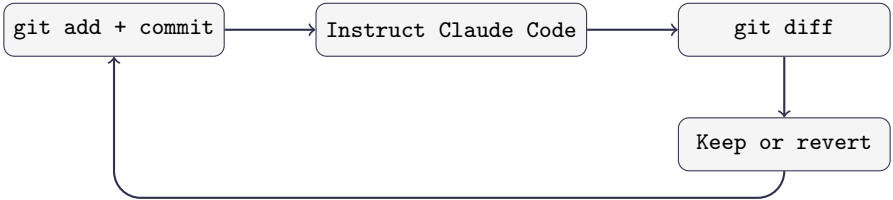
Without git: You try to manually undo changes. You are not sure what the file looked like before. You lose work.

With git: You see exactly what changed, line by line. You revert anything you do not want. You never lose work.

The Rule

Never let Claude Code touch your project without a recent commit to fall back on.

The Minimum Viable Git Workflow



Command	What it does
`git add -A && git commit -m "msg"`	Snapshot your current state
`git diff`	See what changed since last commit
`git checkout -- file.py`	Undo changes to a specific file
`git log --oneline`	See your commit history

Step 1: Commit Before You Instruct

Always save your working state before Claude Code touches anything.

```
git add -A
git commit -m "Working merge script before adding controls"
```

What this gives you:

- A snapshot you can always return to
- A clear "before" to compare against
- Permission to let Claude Code experiment — you can always undo

The Mindset Shift

Commits are not milestones. They are save points. Commit early, commit often, commit before every interaction with Claude Code.

Step 2: Review the Diff

After Claude Code makes changes, see exactly what it did:

```
git diff
```

What to look for:

- Files you did not ask it to change
- Lines removed that should not have been
- "Helpful" additions you did not request

You can also ask Claude Code to explain the diff:

```
Run git diff and explain each change you made.
```

Read the diff, not just the output. The diff shows you what actually changed.

Step 3: Keep or Revert

If everything looks good:

```
git add -A
git commit -m "Add demographic controls"
```

If something is wrong — revert a specific file:

```
git checkout -- scripts/analysis.py
```

If everything is wrong — revert all changes:

```
git checkout -- .
```

The 30-Second Safety Net

This entire cycle takes 30 seconds of overhead. That is the cost of never losing work.

Exercise Part 1: Initialize and Commit (10 min)

1. **Initialize git** in your project directory (if you have not already):
   - `git init`
   - `git add -A && git commit -m "Initial commit"`
2. **Commit** your current work from Sessions 1–2.
   - `git add -A && git commit -m "Session 2 work: merge, estimation, robustness"`
3. **Run** `git log --oneline` to see your history.

You now have a snapshot of everything that works. Whatever happens next, you can always come back here.

Exercise Part 2: The Cycle (25 min)

Do this **at least three times:**

1. **Commit** your current state
2. **Instruct Claude Code** to make a change:
   - Round 1: Add a new control variable to your specification
   - Round 2: Refactor the analysis script into functions
   - Round 3: Change the sample restriction
3. **Run** `git diff` and read what changed
4. **Decide:** keep (commit) or revert (`git checkout -- .`)

Try to revert at least once. The point is to see that you can always undo.

Commit first, instruct second, diff third, decide fourth.

If you take one thing from this module, it is this sequence.

Git Debrief

Discussion:

- Did Claude Code change any files you did not expect?
- Was the diff easy to read? What was confusing?
- Did anyone revert a change? What went wrong?

The Habit

Commit → Instruct → Diff → Decide.

This is not optional when working with an agent that modifies your files. It is the minimum responsible workflow.

In Session 4, we will automate the commit step with a hook — so you never forget.

What if your paper lived in the same place as your code and data?

This is the conceptual pivot of the course.

The Problem with Overleaf

Most researchers keep their paper in a separate tool.

- Code lives in a project directory (or on your Desktop)
- Data lives next to the code (maybe)
- The paper lives in Overleaf (or Word, or Google Docs)

What this means for Claude Code:

- It cannot read your paper
- It cannot check the methods section against your code
- It cannot update a table when the analysis changes
- It cannot commit changes to your paper alongside code changes

The Limitation

If Claude Code cannot see it, Claude Code cannot operate on it.
Your paper is the biggest piece of your project that is currently invisible.

The Solution: LaTeX in VS Code

Bring your paper into the project directory.

- Your `.tex` file lives alongside `scripts/` and `data/`
- Claude Code can read it, write it, and edit it — just like any other file
- Changes to the paper are tracked by git, just like code changes
- Coauthors review changes through version control, not "track changes"

What becomes possible:

- Claude Code writes a methods section and you verify it against the code
- The analysis changes $\rightarrow$ the table updates $\rightarrow$ the paper updates
- Every version of the paper is recoverable through git

The Point

This is not about LaTeX vs. Word. It is about bringing your paper into the agentic workflow.

LaTeX in VS Code: The Setup

What you need (should be installed from pre-work):

1. A TeX distribution: TeX Live (Linux/Windows) or MacTeX (Mac)
2. The **LaTeX Workshop** extension in VS Code

What you get:

- Syntax highlighting for `.tex` files
- One-click compilation to PDF
- PDF preview in a split pane — side by side with your code
- Error messages in the VS Code terminal

The workflow:
Edit `.tex` → Save → Auto-compile → PDF updates in the preview pane.

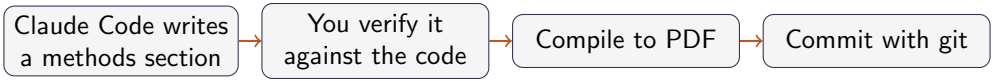
You never leave VS Code. Your paper is right next to your code.

LaTeX Basics (For Those New to It)

```
\documentclass{article}
\usepackage{booktabs}
\title{My Research Paper}
\author{Your Name}
\begin{document}
\maketitle
\section{Data Description}
We use a survey of 2,044 Polish adults...
\section{Results}
\input{output/robustness_table.tex}
\end{document}
```

Key idea: `\input{output/robustness_table.tex}` pulls in the table from Session 2. When the table updates, the paper updates.

Demo: The Full Loop



Everything happens in VS Code.

Everything is version-controlled.

Everything is visible to Claude Code.

Exercise Part 3: Build a LaTeX Document (30 min)

Using the provided template:

1. **Commit** your current state (the habit starts now)
2. **Ask Claude Code** to create a `.tex` file with:
   - A data description section based on your Session 2 pipeline
   - The robustness table via `\input{output/robustness_table.tex}`
3. **Compile** to PDF and verify it renders correctly
4. **Diff and commit** the new file
5. **Ask Claude Code** to write a results paragraph interpreting the table. Review what it writes — would you sign off on this for a paper?
6. **Ask Claude Code** to check the data description against the actual cleaning code. Does the text match what the code does?

Notice: you are now using git and LaTeX together. Every change to the paper is tracked.

What You Built Today

1. **Git:** You have a safety net. Commit → instruct → diff → decide. You practiced reverting changes and saw that you can always go back.
2. **LaTeX in VS Code:** Your paper is now part of your project. Claude Code can see it, edit it, and check it against your code.
3. **The two together:** Every change to your paper is version-controlled, just like every change to your code. You used the git cycle on your `.tex` file.

The Bigger Picture

Your project directory now contains: data, code, output, and a paper — all under version control, all visible to Claude Code. This is the foundation for everything in Session 4.

Debrief

Discussion questions:

1. Did Claude Code change any files you did not expect? How did you handle it?
2. Did anyone revert a change? What went wrong?
3. When Claude Code wrote parts of the LaTeX document, what did you have to correct?
4. How does this compare to your current workflow — Overleaf, Word, separate directories?

Key Takeaway

The skill is not just using Claude Code. It is building an environment where Claude Code can do its best work — and where you can verify and undo everything it does.

Next Session

Session 4: Power Features — Hooks, MCPs, and Skills

- **Hooks:** Automate the commit step — every change Claude Code makes is tracked automatically
- **MCPs:** Connect Zotero — search your library and pull citations from inside Claude Code
- **Skills:** Reusable workflows — spec validation, robustness checks, and more

Before next session:

- Practice the commit-instruct-diff-decide cycle on your own project
- Try asking Claude Code to review a piece of your own code
- Make sure your Zotero account is set up (instructions in the setup guide)